

# AI\_A1\_G11 - Test Log

Generated by running the real commands on this machine. Time: 2026-10-01 12:06:50 South Sudan Standard Time. Python 3.12.3 on Windows-11-10.0.26200-SP0. Git commit at test time: 79e84bf087eee202e10692ee2d912a974a8e28ff

| Test                                                                                                | Exit code | Seconds | Result |
|-----------------------------------------------------------------------------------------------------|-----------|---------|--------|
| 1 Clean full pipeline run                                                                           | 0         | 9.6     | PASS   |
| 2 Predict - valid record                                                                            | 0         | 4.5     | PASS   |
| 3 Predict - incomplete record (must be rejected, exit 2)                                            | 2         | 0.9     | PASS   |
| 4 Predict - malformed JSON (must be rejected, exit 2)                                               | 2         | 0.9     | PASS   |
| 5 Unit tests (12 checks incl. hidden-like data, live-change options, divergence, closed-form check) | 0         | 47.1    | PASS   |

**1 Clean full pipeline run**

```
$ python run_all.py --data data/AI_A1_G11.csv --output artifacts/ --group AI-G11
Group: AI-G11
Rows: 20
Features: 6
SHA-256: 288446a93592327e4c419696e452e0e843882a0e9c08f9dbf6c7531f6eb12e5f
Artifacts generated: classification_metrics.json, cluster_plot.png, clustering_metrics.json, clusters.csv, confusion...
```

**2 Predict - valid record**

```
$ python predict.py --record '{"plot_area_ha": 1.2, "rainfall_mm": 81, "soil_ph": 5.7, "seed_kg": 210, "distance_km": 3195.182671019472, "classification_prediction": 0, "classification_probability": 0.0341129}'
```

**3 Predict - incomplete record (must be rejected, exit 2)**

```
$ python predict.py --record '{"plot_area_ha":1.2}'
{"error": "Missing field(s): rainfall_mm, soil_ph, seed_kg, distance_km, arrival_hour"}
```

**4 Predict - malformed JSON (must be rejected, exit 2)**

```
$ python predict.py --record 'not json'
{"error": "Expecting value: line 1 column 1 (char 0)"} 
```

**5 Unit tests (12 checks incl. hidden-like data, live-change options, divergence, closed-form check)**

```
$ python -m unittest discover -s tests -v
test_01_pipeline_runs (test_pipeline.Pipeline.test_01_pipeline_runs) ... ok
test_02_all_artifacts (test_pipeline.Pipeline.test_02_all_artifacts) ... ok
test_03_hash_matches (test_pipeline.Pipeline.test_03_hash_matches) ... ok
test_04_regression_metrics_recomputed (test_pipeline.Pipeline.test_04_regression_metrics_recomputed) ... ok
test_05_confusion_matrix_consistent (test_pipeline.Pipeline.test_05_confusion_matrix_consistent) ... ok
test_06_every_record_clustered (test_pipeline.Pipeline.test_06_every_record_clustered) ... ok
test_07_predict_valid (test_pipeline.Pipeline.test_07_predict_valid) ... ok
test_08_predict_rejects_bad_input (test_pipeline.Pipeline.test_08_predict_rejects_bad_input) ... ok
test_09_hidden_like_data (test_pipeline.Pipeline.test_09_hidden_like_data) ... ok
test_10_live_change_options (test_pipeline.Pipeline.test_10_live_change_options) ... ok
test_11_diverging_lr_gives_clear_error (test_pipeline.Pipeline.test_11_diverging_lr_gives_clear_error) ... ok
test_12_gradient_descent_matches_closed_form (test_pipeline.Pipeline.test_12_gradient_descent_matches_closed_form) .

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Ran 12 tests in 45.353s

OK
```